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THE INTERACTION OF HYDROGEN WITH A RHODIUM (110) SURFACE

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The dissociative adsorption of hydrogen and deuterium on a Rh(110) surface was studied in UHV at 80 K by means of LEED, thermal desorption spectroscopy (TDS) and work function ($\Delta\varphi$) measurements. We find atomic desorption in three different binding states α_1, α_2 and β having activation energies for desorption of ~33, ~53 and ~90 kJ/mol, respectively. The initial sticking probability, s_0, is 0.97 ± 0.10; the adsorption kinetics follows a precursor state model. The saturation coverage reaches the abnormally high value of 2 H atoms per Rh surface atom. Hydrogen forms a variety of ordered lattice gas phases, namely, a p(1×3) at $\theta_H = 0.33$, a p(1×2) at $\theta_H = 0.5$, a 1×3-2H at $\theta_H = 0.67$, a 1×2-3H at $\theta_H = 1.5$ and a 1×1-2H at $\theta_H = 2.0$ with pronounced lateral interaction energies between the adsorbed H atoms. A layer relaxation or slight surface reconstruction induced by adsorbed H cannot be ruled out from our data.

1. Introduction

The interaction of hydrogen with metal surfaces is a matter of widespread interest since many years, for a number of reasons [1,2]. Owing to its small size the H atom can easily induce displacements of surface atoms (the so-called "H-induced surface reconstruction") as was shown, e.g., for the systems H/Ni(110) [3] and H/Pd(110) [4]. It appears as if the crystallographically "open" surfaces (like fcc (110) or hcp ($10\bar{1}0$)) were particularly susceptible to this kind of chemical attack which could be regarded as one of the first steps towards a compound formation (metal hydride). In this respect it is also believed that an understanding of these fundamental processes could provide insight into the still open and challenging problems of the H-induced metal embrittlement and crack phenomena [5].

Rhodium is known to be a very active catalyst material and is thus used for industrial applications. It can reduce NO_x compounds to molecular nitrogen and is therefore used as an active component in the three-way catalysts in automobiles. Moreover, rhodium and its compounds play an important role in

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the organo-metallic chemistry as homogeneous catalysts. Nevertheless only few basic studies concerning the adsorptive properties of well-defined Rh surfaces have been carried out so far, compared to Ni, Pd or Pt surfaces. The interaction between hydrogen and rhodium has only quite superficially been investigated in the past: There are early works by Beeck [6] and Mimeault and Hansen [7] on Rh films. Castner et al. [8] studied the adsorption of hydrogen (and a variety of other molecules) on Rh(100) and Rh(111) surfaces by means of LEED and thermal desorption spectroscopy. Later, Yates, Thiel and Weinberg [9] performed a more extensive study of hydrogen chemisorption on Rh(111), but no other detailed investigation of the interaction of hydrogen with Rh single crystal surfaces, in particular the Rh(110) surface, has come to our attention. Several groups studied, of course, polycrystalline Rh surfaces (films, foils, wires) and their interaction with hydrogen and other gases, for example Wells et al. [10] or Edwards et al. [11], mostly for catalytic purposes. Also, Rh surfaces have been studied in the context of supported catalysts [12] or bimetallic catalysts in conjunction with deposited silver atoms [13] but there is a lack of primary information of the interaction of Rh surfaces with hydrogen. It was therefore attempted in this work to collect detailed information about thermodynamic and kinetic properties (heats of adsorption, activation energies for desorption, sticking probabilities, frequency factors) as well as structural data in order to compare the Rh(110)/H system with its congeners in Group VIIIb of the periodic table, Ni(110)/H [14,15] and Pd(110)/H which all crystallize in the face-centered cubic system.

Some of the most striking results, namely, the formation of true hydrogen LEED superstructures with pronounced long-range order as well as the high concentration of adsorbed hydrogen under saturation conditions (the coverage $\theta_H = 2$ ML (monolayers) is attainable) have already been published separately [16]. Here we give a more extensive description of our findings about H/Rh(110).

2. Experimental set-up and procedure

The experiments were carried out in an all stainless-steel chamber pumped by a turbo-molecular pump and equipped with a 4-grid LEED optics containing a moveable Faraday cup, a quadrupole mass spectrometer (Balzers, QMG 311) inside a differentially pumped cone and a home-made vibrating Kelvin device for work function measurements. Besides that it contained the standard facilities (argon sputter gun, Auger electron gun) for cleaning the surface and characterizing the chemical surface composition. The crystal was of 5N purity and was purchased from Technisches Büro für Kirstallzüchtung G. Lamprecht. Before mounting in vacuo the crystal was X-ray oriented and mechanically polished. Opposite to many other Rh samples it did not contain boron, but

rather sulfur and carbon as major impurities. Prolonged cycles of Ar ion sputtering, followed by annealing at 1000–1200 K and subsequent mild oxygen and hydrogen oxidation and reduction cycles, respectively, finally led to a chemically very clean and crystallographically well oriented sample. It could be cooled to ~ 85 K by means of a differentially pumped liquid nitrogen cold finger and heated by a programmable dc heating unit so as to allow temperature programmed thermal desorption experiments with a linear heating rate of 10 K s^{-1}. The Rh crystal was spot-welded to two parallel running Ta wires which in turn were attached to 2 mm diameter W rods; a chromel–alumel thermocouple served for the temperature measurements.

During the TDS experiments the sample was turned directly in front of the aforementioned differentially pumped cone. Thus, contributions of desorbed species from the sample holder were minimized. Work function data were collected with an accuracy of better than ±1 meV with a Kelvin probe which consisted of a magnetically driven glass arm with a gold ring electrode on its one end. The working frequency was 230 s^{-1}, the long-term stability was better than 5 meV/h.

The gases (Linde, Messer Griesheim) were admitted via all-metal bakeable leak valves (Varian), the gas forelines were liquid nitrogen cooled in order to remove water vapor which turned out to be a very awkward impurity since below 150 K it adsorbs competitively with high sticking probability on the Rh surface.

An excellent monitor to probe coadsorption of water was the work function change which increases by ~ 930 meV at saturation if pure H_2 is admitted but decreases by more than 1 eV, if the Rh surface is exposed to water vapor instead of hydrogen. The pressures and exposures given in this work are corrected with respect to the ion gauge factor; for H_2 and D_2 we have chosen a sensitivity factor of 0.5.

3. Results

3.1. Low energy electron diffraction (LEED)

The clean and well-annealed Rh(110) surface gave rise to a LEED pattern with bright and sharp 1×1 spots on a negligible background, if the surface was cooled to ~ 100 K. The ultimate pressure of the chamber had to be kept below 10^{-10} mbar in order to avoid spurious adsorption of CO, H_2O or H_2. We want to note here that a more extensive description of the H-induced LEED features including spot profiles will be reported elsewhere [17]; here we only give a summary of the observed phenomena. Exposure to hydrogen causes a whole variety of ordered LEED structures which exhibit continuous order–disorder transitions and all have fairly weak intensity in the "extra"

spots. The first structure appears on the cold surface ($T \leq 100$ K) already after very short exposures to H_2 (~ 0.1 L) which indicates a markedly high sticking probability of hydrogen on Rh(110). It is a p(1 × 3) structure with sharp, round but weak "extra" spots, the intensity being less than 1/10 of that of the substrate lattice spots. They reach maximum intensity after 0.11 L (D_2: after 0.16 L) and then become continuously elongated in [001] direction until the region between the $(\frac{1}{2}, 0)$ and $(\frac{2}{3}, 0)$ spots is uniformly filled with scattered intensity. Continuous exposure then causes the appearance of a single maximum of a p(1 × 2) structure. This 1 × 2 phase reaches maximum intensity after ~ 0.19 L (0.26 L for D_2). Again, the "extra" intensity is very weak. Further exposure leads again to a broadening of the H-induced spots in [001] direction until after 0.29 L (0.4 L for D_2) a "second" 1 × 3 phase appears. To obtain this phase requires careful dosing and a slight annealing of the exposed sample to ~ 120 K; even then the "extra" spots are not exactly "in place", that is, in $(\frac{1}{3}, 0)$ and $(\frac{2}{3}, 0)$ positions, but closer together, perhaps in $(\frac{3}{8}, 0)$ and $(\frac{5}{8}, 0)$ positions. However, if the phase is generated by cooling a sample from higher temperature to below 150 K the spots move to the exact positions of a 1 × 3 phase. It appears therefore as if the complete order requires the (thermally activated) rearrangement of a certain stable domain size distribution. As the H_2 exposure is further increased the 1 × 3 spots gradually become weaker and transform to a second 1 × 2 phase which reaches maximum intensity around 1.1 L (1 L with D_2). The intensity of this 1 × 2 phase (which is still weak) then decreases continuously as the H_2 exposure increases further and finally disappears completely, provided the sample temperature is always kept below 100 K and the LEED beam is turned on only for a short period of time. The finally formed 1 × 1-H phase seems to be completed after approximately 5–6 L H_2 or D_2. Already short electron beam impact makes the 1 × 2 phase reappear because of electron beam induced desorption of particles from the 1 × 1-2H phase.

It should be mentioned that the LEED phases generated by dosing with *deuterium* instead of hydrogen showed a better two-dimensional ordering (somewhat brighter "extra" spots, lower background intensity). Although the overall weak intensity of the H induced LEED features suggests that the Rh surface does probably not undergo a marked surface reconstruction we have followed the intensity–voltage (I–V) curves of some integer-order beams, namely, the (0, 0) and the (0, −1) beam, with our Faraday-cup assembly and could in neither case detect any *significant* alterations of the I–V curves due to adsorbed hydrogen except a shift of the Bragg maxima by about 1 eV to lower energies. More systematic and quantitative I–V measurements using a fast Video–LEED system have been performed very recently [18].

The major result of these experiments is that in the final 1 × 1-2H phase the adsorbed H atoms remove the inherent layer relaxation of the topmost Rh atoms completely. This means that the first interlayer spacing which is about

7% contracted for the undercovered surface increases by 5.5% and the 1.9% second layer expansion of the clean surface is reduced by about 1.7% as the final 1×1-2H phase forms. Moreover there is some evidence that even in the first p(1×2) phase at $\theta_H = 0.5$ a slight surface reconstruction is involved, since the "extra" intensity is in general comparatively high and, more importantly, the "extra" spots remain sharp and visible up to more than 300 eV electron energy which is not to be expected for mere H scattering. This could be understood in terms of a (slight) up–down reconstruction of the topmost Rh atoms. Recent HREELS measurements by Conrad et al. [19] support this view in that Conrad et al. report on a pronounced change in the electron surface reflectivity even at fairly small overall H coverages well below 1 monolayer, which could be caused by a slight additional roughening of the Rh(110) surface due to this up–down reconstruction (but also be caused by changes of electronic surface states by H adsorption). Definite conclusions, however, can only be reached from a fully dynamical structure analysis which is currently in progress [20].

In order to investigate the thermal stability of the various H(D)-induced LEED phases we have generated the respective structures and followed the intensity–temperature dependence. Details of these experiments will be described elsewhere [17]; here we only briefly report on the most striking phenomena. The $I(T)$ measurements work quite well for the first three ordered phases (which consist exclusively of of strongly chemisorbed hydrogen). From the inflection point of the respective LEED intensity versus temperature curves (which all indicate continuous order–disorder phase transitions) critical temperatures of ~ 165, ~ 170 and 210 K (all temperatures accurate within 10 K) can be derived for the p(1×3), p(1×2) and 1×3 phases, respectively. Because of competing thermal desorption it was not possible to follow, at constant coverage θ, the order–disorder transition of the 1×2 and 1×1-H phases. Upon even slight heating the respective phases disappear irreversibly.

3.2. Thermal desorption spectroscopy

3.2.1. The hydrogen binding states

Fig. 1 shows a series of thermal desorption spectra obtained with a heating rate of $\beta = 10$ K s^{-1} after the Rh(110) sample had been exposed to hydrogen at ~ 85 K. Three different binding states are populated subsequently, a β state, an α_2 state and a (comparatively narrow) α_1 state. The high temperature β binding state becomes filled after ~ 0.40 L exposure. It exhibits the typical shift of the desorption maximum towards lower temperatures as the coverage is increased, i.e., from ~ 280 K for the zero coverage limit to ~ 245 K at saturation. This is indicative of second-order kinetics. Next the α_2 state at $T_{max} = 216$ K is populated until it saturates after roughly 1 L exposure. Here,

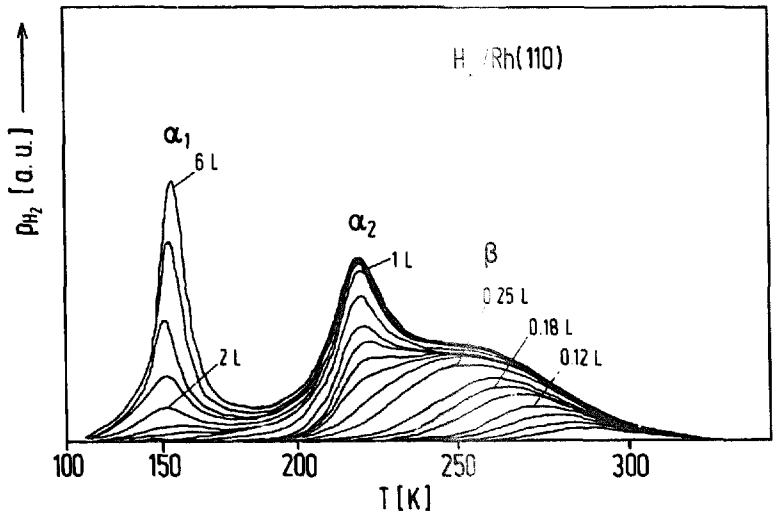


Fig. 1. Series of thermal desorption (TD) spectra from a Rh(110) surface exposed at 85 K to increasing amounts of H_2 gas. The various H binding states are indicated.

practically no peak shift is observed (first-order desorption). Thereafter the low temperature α state appears around 138 K whose maximum is again invariant with coverage. It is completed after an exposure of ~ 6 L. It should be emphasized that H_2–D_2 exchange experiments reveal clearly the atomic nature of all three binding states: If the Rh(110) surface is subjected to a mixed exposure, i.e. 50% H_2 and 50% D_2, all three states can also be observed with mass 3, the HD compound.

An evaluation of the activation energy for desorption seems to be straightforward for the β state for which we can assume a true second-order rate process and an approximate coverage-independent heat of adsorption.

According to the simple Redhead equation for second-order desorption [22]

$$\frac{E_{\mathrm{des}}}{RT_{\mathrm{max}}^2} = 2\sigma_{\mathrm{max}}\frac{\nu_2}{\beta}\exp\left(-\frac{E_{\mathrm{des}}}{RT_{\mathrm{max}}}\right) \tag{1}$$

(in which E_{des} is the activation energy for desorption, σ_{max} the surface coverage at peak maximum, ν_2 the second-order pre-exponential, β the heating rate, R the gas constant, and T_{max} the temperature at which the desorption rate peaks) which can be rearranged to (with $\sigma_0 = 2\sigma_{\mathrm{max}}$ = TD peak area):

$$\ln\left(\sigma_0 T_{\mathrm{max}}^2\right) = \frac{E_{\mathrm{des}}}{RT_{\mathrm{max}}} + \ln\left(\frac{\beta E_{\mathrm{des}}}{\nu_2 R}\right) \tag{2}$$

a plot of $\ln(\sigma_0 T_{\mathrm{max}}^2)$ versus $1/T_{\mathrm{max}}$ should yield a straight line of slope E_{des}/R. Fig. 2 shows this plot for the β_2 state. It is evident from the plot (despite the

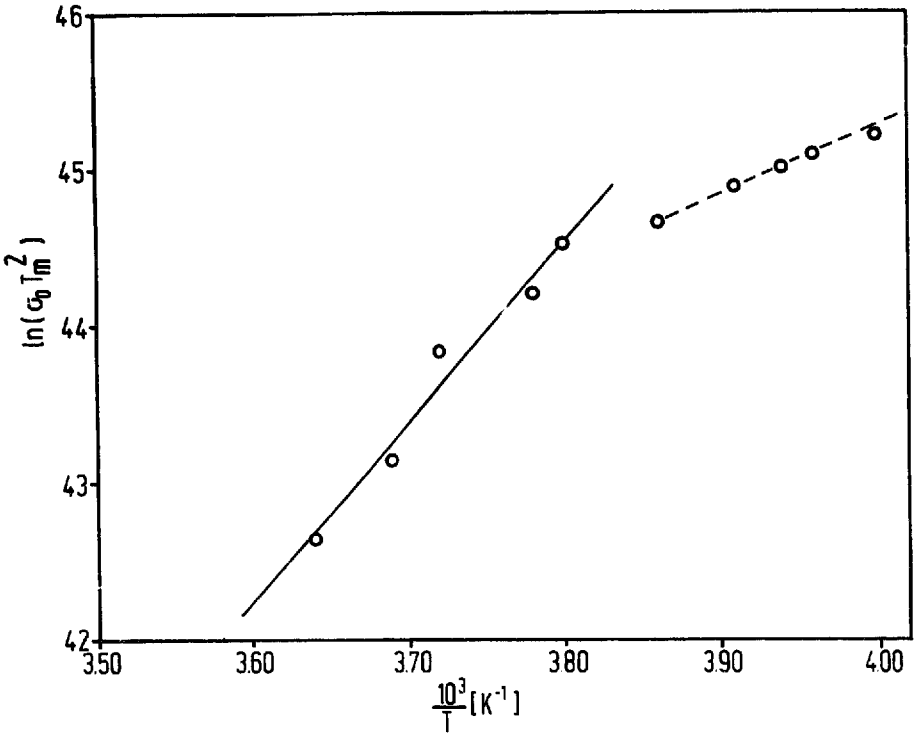


Fig. 2. Second-order evaluation according to Redhead [22] for the H β state desorbing from Rh(110).

scatter of the data) that two regions can be distinguished: a low coverage region up to 0.25 L in which a straight line can be fitted to the data, and a high coverage region in which the data again seem to obey a linear relationship. However, since these data may be influenced by contributions of the α_2 peak only the first sequence of data is analyzed according to the Redhead formalism. We obtain a coverage-independent activation energy for desorption of ~ 90 kJ/mol for the β_2 state which is accurate to within 5–10 kJ/mol. This value is in fairly good agreement with results obtained from isosteric heat measurements to be reported in section 3.3.2. From these measurements it also turns out that the assumption of a coverage-independent heat of adsorption is probably correct up to a hydrogen coverage of at least 0.6 monolayers. However, if there will be changes in the frequency factor (which is feasible because of the structural changes in the adsorbed layer) the E_d^* evaluation could be noticeably influenced.

The activation energy for the α_2 state can be derived under the assumption of a first-order desorption process whose activation energy is coverage-independent. This is justified by the almost constant position of the α_2 peak at 216 K. As was demonstrated by Redhead [22] also the first-order desorption

process can be described by a good approximation (provided E_{des} does not depend on coverage):

$$E_{des} = RT_{max}[\ln(\nu_1 T_{max}/\beta) - 3.64], \quad (3)$$

which yields in the present case $E_{des,\alpha_2} = 53$ kJ/mol, if ν_1 is commonly taken as 10^{13} s^{-1}.

Likewise, an *approximate* value for the desorption energy of the α_1 state may be deduced in this way, because its peak shift is sufficiently small. We obtain, with $T_{max} = 138$ K and $\nu_1 = 10^{13}$ s^{-1}, the strikingly low value of $E_{des,\alpha_1} = 33.2$ kJ/mol. It is important here to point out the close correlation between the appearance of the H-induced LEED structures and the formation of the various desorption states: Whilst the β state contains the first three H phases (p(1 × 3), p(1 × 2) and 1 × 3) the α_2 state is associated with the formation of the second 1 × 2 phase (in which the H atoms experience a significantly higher packing density) and the α_1 state is most pronounced when the final 1 × 1-2H phase has reached saturation. In this phase the adsorbate density is once more appreciably higher than in all other phases. Accordingly (and this will be made clear in the discussion (section 4) with the aid of the structure models for the ordered H phases) the origin of the α_2 and α_1 state can be traced back to the operation of strong repulsive mutual hydrogen interactions.

Thermal desorption spectroscopy has proven to be a very useful tool to determine at least relative adsorbate coverages: The area under a TD peak reflects directly the adsorbed amount in this state at the desorption temperature. Therefore we can, in a first step, integrate those TD curves which correspond to the full development of the various lattice gas LEED structures in order to obtain the relative coverages associated with those phases. The results are compiled in table 1 (we have taken the first p(1 × 3) structure as a coverage reference: Its intensity maximum should reflect just a H coverage of 1/3 of a monolayer, i.e., 3.27×10^{14} H atoms/cm^2).

Apparently, within the limits of accuracy, the first 1 × 2 phase contains a coverage of 1/2, the second 1 × 3 phase $\theta_H = 2/3$, the second 1 × 2 phase $\theta_H = 3/2$ and the final phase $\theta_H = 2$, whereby the coverage θ_H is defined as

Table 1
Summary of the H LEED structures, exposures and coverages

LEED structure	Exposure (L)		Relative coverage
	H_2	D_2	
p(1×3)	0.12	0.16	0.33 = 1/3
p(1×2)	0.18	0.26	0.53
1×3	0.28	0.40	0.69
1×2	~1	1.0	1.47
1×1	~5	5	1.97

n/n_{max}, with $n_{max} = 9.8 \times 10^{14}$ Rh atoms/cm^2. Not only provides the knowledge of these coverages us with a handle to derive reasonable structure models to explain the LEED structures (as is shown in the discussion (section 4)) but we can also directly determine the relative and absolute sticking probability of hydrogen as a function of the H coverage.

3.2.2. The kinetics of adsorption

Fig. 3 shows the uptake of the surface with adsorbed hydrogen and deuterium as the exposure is increased. Obviously there is an almost linear initial increase up to $\theta = 0.5$ to 0.6; thereafter the curve declines and slowly bends off. The *relative* coverage data could be obtained from the TD integrals $\int p \, dt$ with fairly good accuracy. The conversion to absolute coverages was again achieved by means of the p(1 × 3) LEED pattern which was set equal to $\theta_H = 0.33$. Here the accuracy is certainly not better than 10%. Nevertheless, the slope of the curve of fig. 3 yields the sticking probability as a function of coverage, $s(\theta)$ which is reproduced in fig. 4. The structure of this curve is correlated with the formation of the various ordered LEED phases: During the formation of the first p(1 × 3) phases s remains practically constant (0.97 ± 0.10) pointing to a precursor state. As the p(1 × 2) phase forms, s decreases to 0.8 and is again constant until the 1 × 3-2H phase is completed. Thereafter, we find again a pronounced drop associated with the formation of the 1 × 2-3H phase beginning around $\theta_H = 0.9$. Surprisingly, this drop is much more pronounced with hydrogen than with deuterium (H: open circles and full line; D: full circles and dashed line). As the 1 × 2-3H phase is saturated there

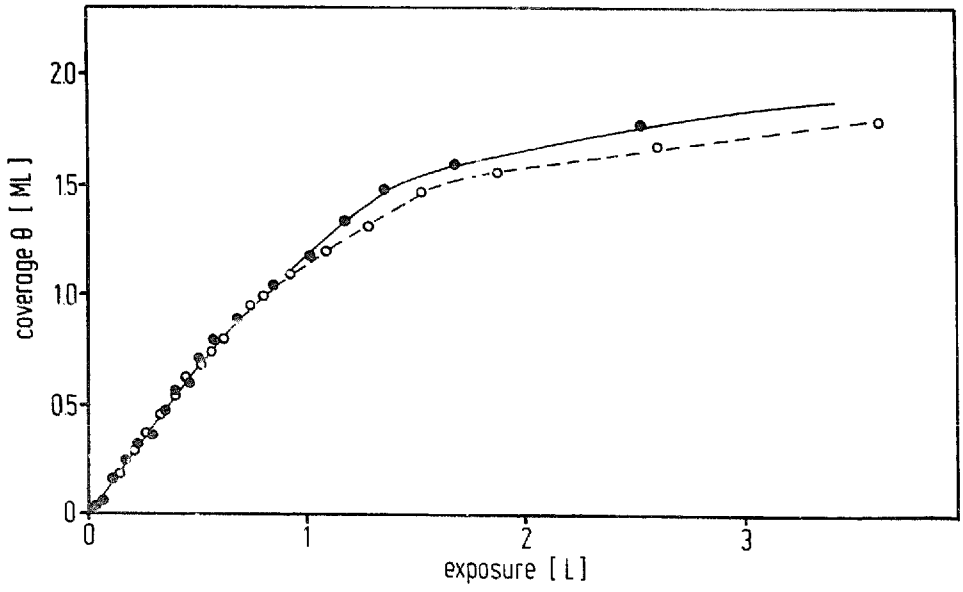


Fig. 3. Hydrogen (dotted line) and deuterium (full line) uptake of the Rh(110) surface at ~ 85 K as a function of exposure as measured by the TDS integral $\int p \, dt$, at $T = 85$ K.

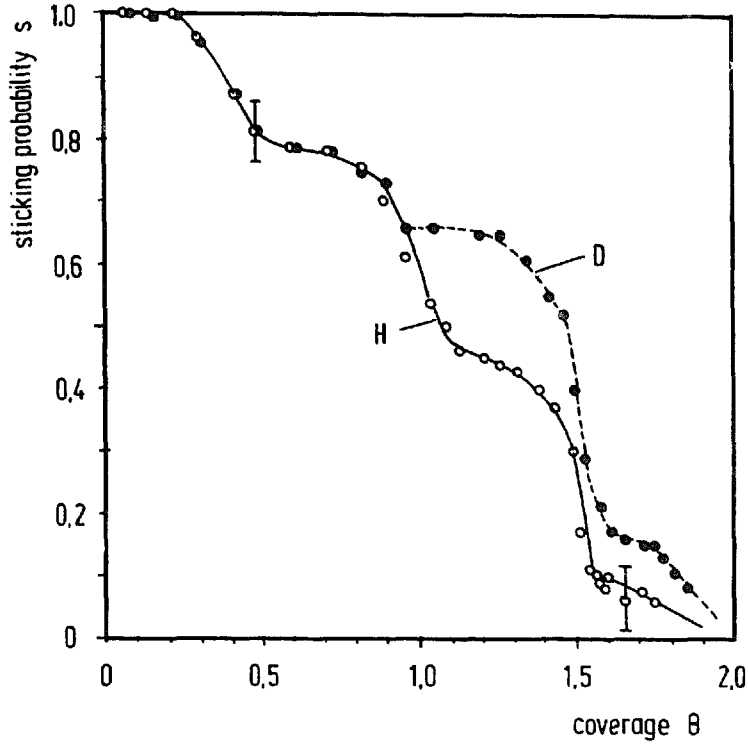


Fig. 4. Sticking probability, s, as a function of hydrogen (deuterium) coverage, at $T = 85$ K. The convex shape of the curve suggests a precursor state adsorption kinetics.

is another little bump which is correlated with the formation of the final 1×1-2H phase. Again, a slight difference between the sticking probabilities of H and D becomes visible. A detailed discussion of the isotope effect of the H(D)/Rh(110) system is given elsewhere [23].

3.3. *Work function ($\Delta\varphi$) measurements*

3.3.1. *Surface dipole moments*

The LEED and thermal desorption data are backed up and supplemented by the work function change measurements. If the Rh sample kept at ~ 90 K is exposed to hydrogen the work function increases by about −930 meV at saturation. As fig. 5 illustrates there is by no means a linear relation between $\Delta\varphi$ and the hydrogen exposure, rather a complicated curved relationship arises. Initially there is an almost linear increase up to approximately 100 meV, then the slope *increases* followed by a second linear increase up to ~ 650 meV before again a bend adds another 330 meV to the final work function change of 930 meV after ~ 5 L exposure. It is now quite revealing to compare the exposure associated with the H LEED structures with these different regions of the work function change: The respective exposures are

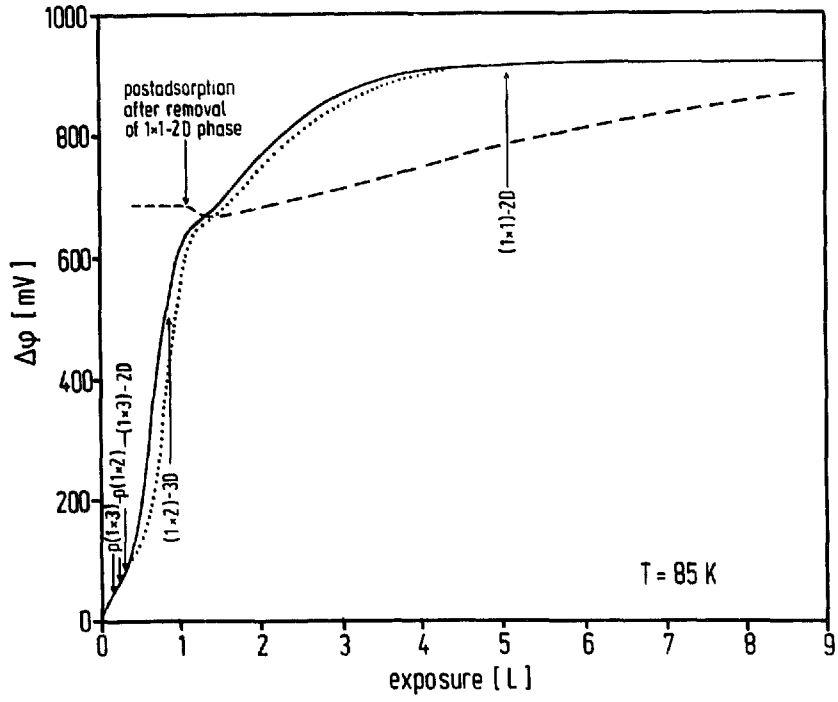


Fig. 5. H-induced work function change as a function of exposure, at $T = 85$ K. The arrows indicate the respective completion of an ordered H phase. The broken line describes the situation observed with deuterium instead of hydrogen.

marked in fig. 5 by arrows. A comparatively small and homogeneous work function contribution is provided by the first three lattice gas structures thus indicating that the dipole moment of the H atoms in these structures is small and of similar character. At these low coverages we may neglect depolarization effects and may therefore use the Helmholtz equation

$$\Delta\varphi = 4\pi\sigma_0\mu_0 f^*, \tag{4}$$

where μ_0 is the initial dipole moment, σ_0 the number of adsorbed H atoms/cm^2, and $f^* = 1/4\pi\epsilon_0$ the conversion factor.

If we take $\sigma_0 = 0.5$ ML $= 4.925 \times 10^{14}$ H atoms/cm^2 and $\Delta\varphi \approx 55$ meV associated with the p(1 × 2) phase we obtain $\mu_0^{p(1\times2)} = 8.99 \times 10^{-32}$ C m = 0.027 debye (D) (1 C m = 3×10^{29} D). This is a very small dipole moment and underlines the covalent character of the Rh–H bond. It also illustrates that the H atoms are probably located very close to the surface, and it rules out *major* structural rearrangements of the Rh surface at this stage of H surface concentration.

The situation changes as the second 1 × 2 phase (1 × 2-3H) forms. $\Delta\varphi$ exhibits an overproportional rise, and the dipole moment associated with the H atoms in the 1 × 2 configuration amounts to $\mu_0^{1\times2} = 3.90 \times 10^{-31}$ C m = 0.12 D, i.e., is almost a factor 5 larger. The actual dipole moment of the individual

H–Rh complex is probably even higher since the H atoms are quite closely packed in the 1×2 phase and will likely exert depolarization effects on their neighbours. Furthermore, also an influence of relaxation effects of the Rh surface atoms cannot be ruled out but rather must be considered to contribute to the overall work function change.

That indeed some peculiar effects are going on as the 1×2-3H phase is built up can be seen from the $\Delta\varphi$–exposure curve, if D_2 is used for adsorption (dotted line). Whereas no difference is observed in the $\Delta\varphi$–exposure curve for the first three phases the formation of the 1×2-3H phase starts earlier, i.e., at a higher rate with D_2 than with H_2. This effect is due to differences in the sticking probability between H_2 and D_2 in this range and has also shown up in the TD spectra of fig. 1. Exactly the same holds for the final branch of the $\Delta\varphi$–exposure curve which is directly correlated with the formation of the 1×1-2H phase. Again the dipole moment $\mu_0^{1\times 1\text{-}2H}$ can be calculated to yield a value of 4.18×10^{-31} C m = 0.13 D which is similar to what has been found for the preceding 1×2 phase. This may indicate a similar configuration and charge transfer in these two high coverage binding states. Another hint at the peculiar character of these phases is provided by the relatively sharp α_2- and α_1-TD states and, of course, by the structure models developed on the basis of

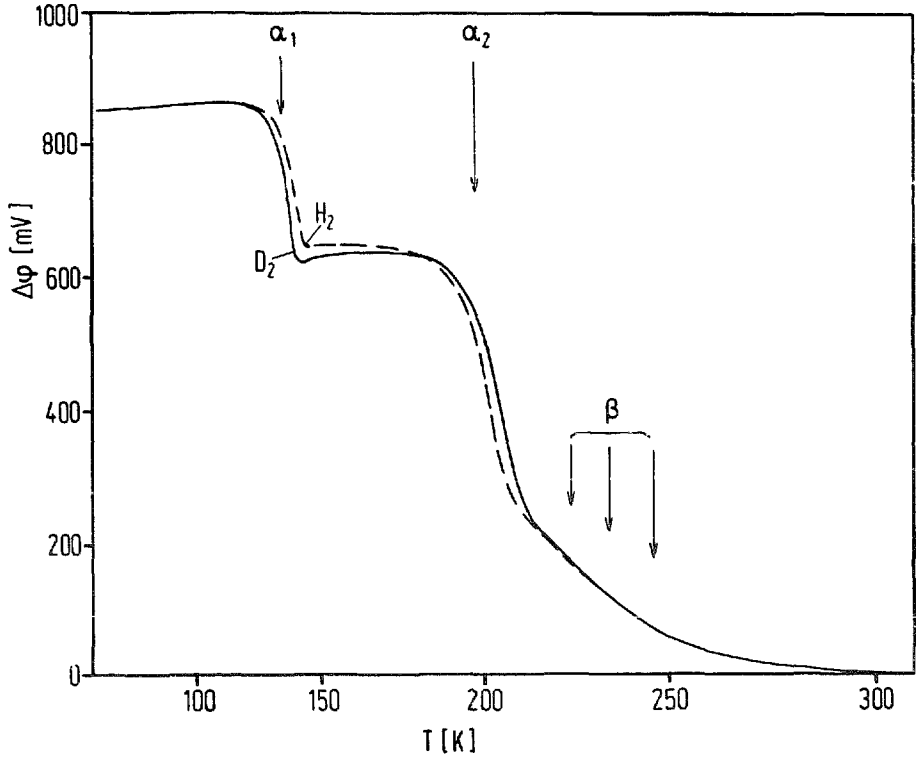


Fig. 6. Temperature dependence of the H(D)-induced work function change, $\Delta\varphi(T)$. The desorption from the various H (D) binding states is marked by arrows.

the LEED features as will be described in the discussion (section 4). The close correlation between the different work function values and the thermal desorption states can nicely be seen if the temperature dependence of the work function change is monitored. Experimentally this was achieved by heating the saturated surface linearly at a rate of 0.3 K s^{-1} while recording the work function change on a x–y recorder. The corresponding $\Delta\varphi(T)$ curve is shown in fig. 6. Clearly, the onset of thermal desorption causes a step-like decrease in the region of the α_1 state and a somewhat shallower but still pronounced fall around 220 K according to desorption from the α_2 state. The desorption from the β state then follows as a much slower process between 240 and 300 K. Taking into account the different dipole moments of the α and β H states we find a clear 1:1 correspondence between partial pressure and work function change measurements.

3.3.2. $\Delta\varphi$ as a coverage monitor

The exposure scale of fig. 5 can be converted into a true coverage scale if one makes use of the known sticking probability–coverage function (cf. figs. 3 and 4). The resulting $\Delta\varphi(\theta)$ relation is shown in fig. 7 which allows a better discussion of the structure dependence of the dipole moments although basically the same features can be distinguished. It turns out, as expected, that the first three phases exhibit similar (small) dipole moments but, after $\theta > 0.7$, a steep $\Delta\varphi$ rise occurs which comes to an end only around $\theta = 1.5$, i.e., when the 1×2-3H structure is completed. Thereafter the $\Delta\varphi(\theta)$ curve bends off before it rises again for $\theta > 1.7$ ML, with the same slope (dipole moment) as the former phase. Finally, around $\theta_H = 1.9$ ML the curve levels off until saturation is achieved at $\theta = 2.0$.

For a not too large coverage range in which the adsorption/desorption equilibrium can be established in the accessible temperature range by reasonable gas phase pressures, i.e., for $0 < \theta_H < 1$, we can determine the coverage (as indicated by $\Delta\varphi$) as a function of temperature and pressure and thus obtain adsorption isobars. Since, however, the overall $\Delta\varphi$ is quite small up to this coverage a good accuracy of the data is required. Moreover, the non-linear $\Delta\varphi(\theta)$ relation has to be considered above $\Delta\varphi = 200$ meV. Provided that there is always equilibrium: gas phase–adsorbate phase we can use the simplified Clausius–Clapeyron equation:

$$\left(\frac{\mathrm{d}\ln P}{\mathrm{d}(1/T)}\right)_\theta = -\frac{q_{st}}{R} \tag{5}$$

to describe the phase equilibrium. In fig. 8 we present the respective adsorption isotherms for H adsorption on Rh(110).

According to eq. (5) a plot of ln P versus T^{-1} should yield straight lines for each individual coverage, the "isosteres", the slope of which contains the

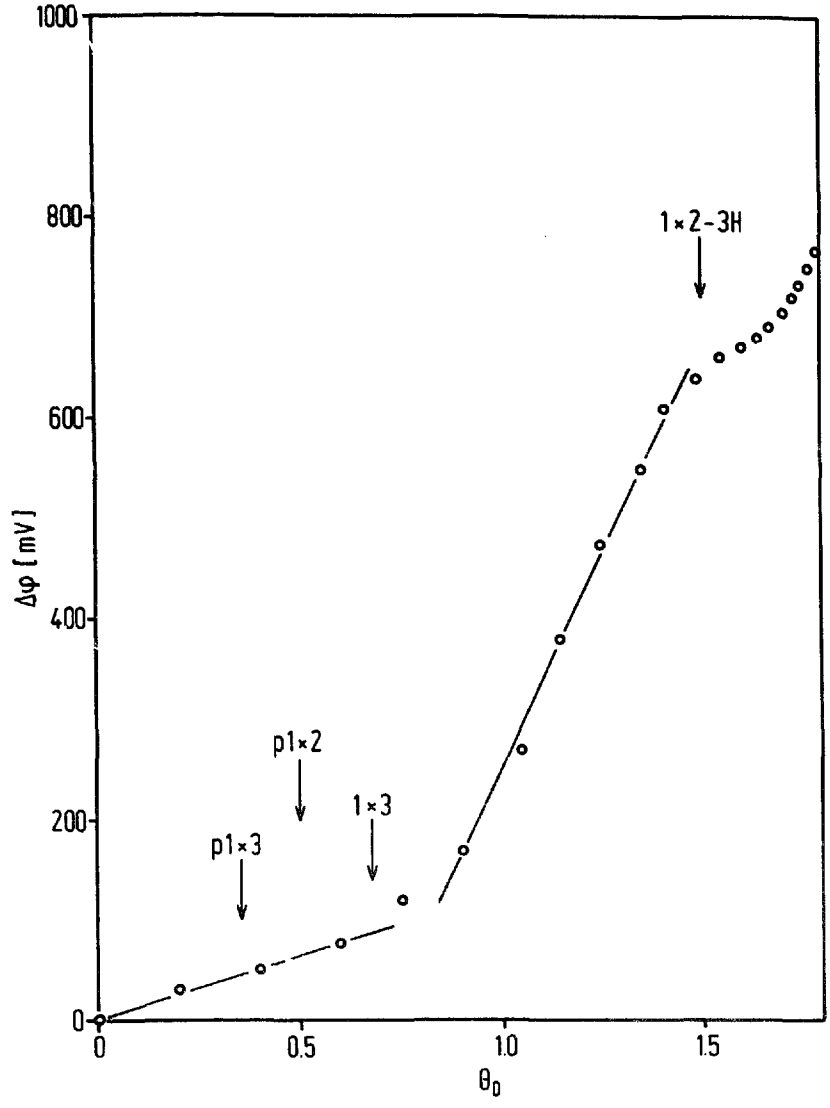


Fig. 7. H-induced work function change as a function of coverage, $\Delta\varphi(\theta)$, at $T = 85$ K. Note the different slopes of the curve which correlate with the existence of the various ordered H phases (see text for details).

isosteric heat of adsorption, q_{st}. From several such isosteres taken at different coverages we obtain the function $q_{st}(\theta)$ which is plotted in fig. 9.

We find for the region $0 < \theta_H < 0.7$ a nearly coverage independent value of q_{st} around 80 kJ/mol, which agrees well with the activation energy for desorption as determined from the Redhead analysis. At $\Delta\varphi \approx 200$ meV which corresponds to $\theta_H = 0.7$) we observe a dramatic fall of q_{st} down to values around 50 kJ/mol. This coincides with the beginning formation of the 1×2-3H phase (i.e., the α_2 state in the TD spectra) for which we estimated the activation energy of desorption to be 53 kJ/mol. Again, we come to the

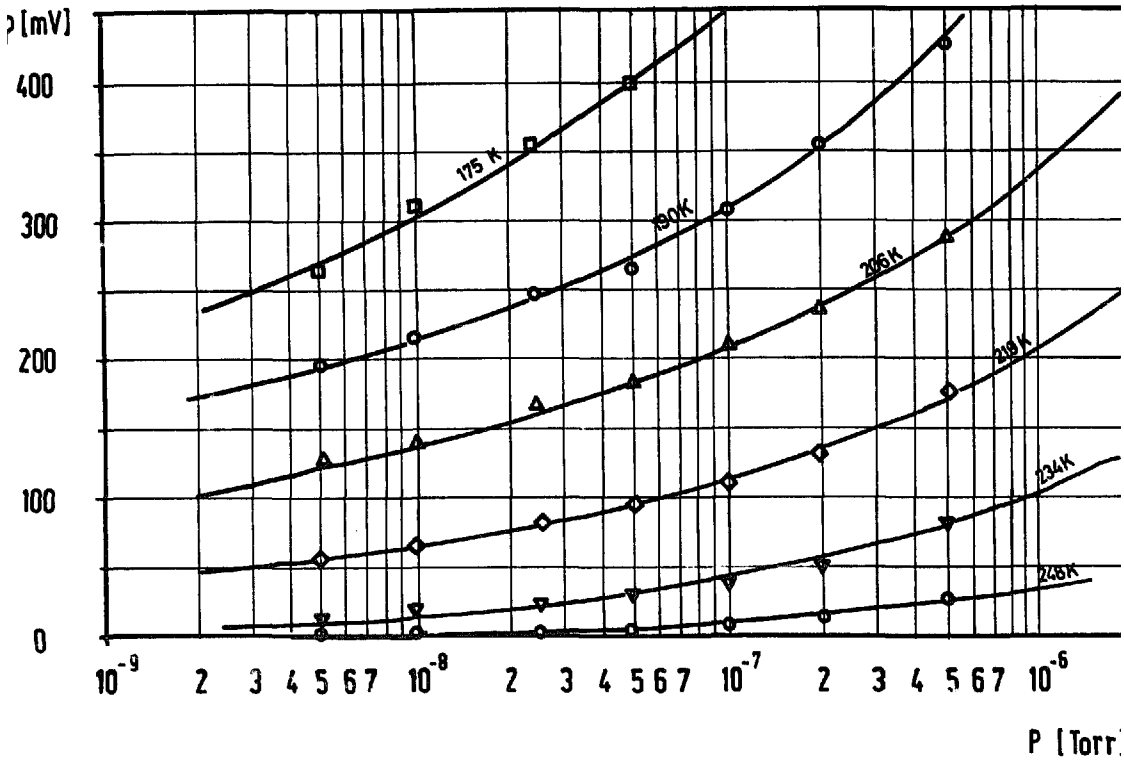


Fig. 8. Adsorption isotherms ($175 < T < 250$ K) for H on Rh(110).

conclusion that this phase exhibits a substantially weakened H–metal bond, due to lateral interactions.

A previous work by Gorodetskii et al. [54] using a Rh field emitter reported 79 kJ/mol for q_{st} of H at low coverages, in remarkable agreement with our result.

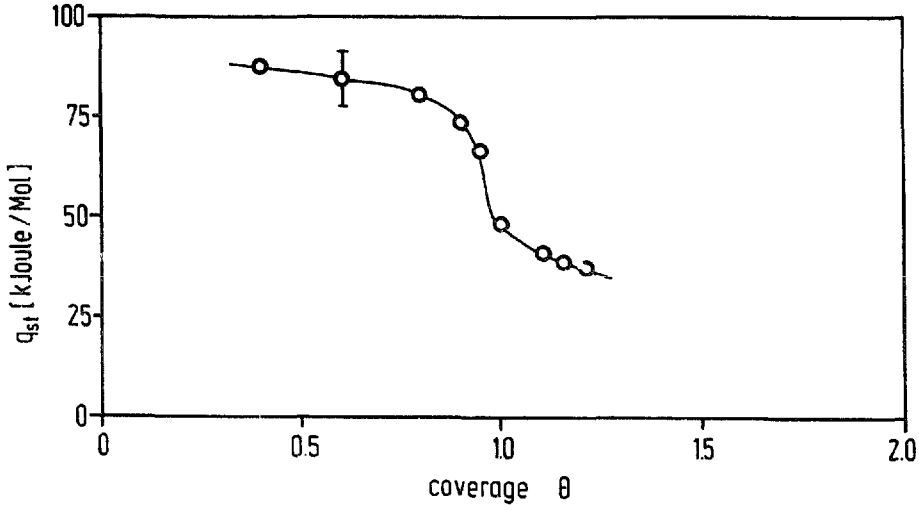


Fig. 9. Isosteric heat of adsorption, q_{st}, as a function of H coverage, for the low coverage limit.

4. Discussion

In this section we focus on two subjects, namely, the structural properties of the H/Rh(110) system as they arise from the surface binding situation and on some aspects of the adsorption/desorption kinetics as probed by our thermal desorption and work function change measurements.

Of particular interest will be a comparison of the H chemisorptive properties of the Rh surface with its neighbours in the periodic table, Ru and Pd, for which recent results exist [24,4].

4.1. The binding states and the structure of the adlayers

The thermal desorption spectra of fig. 1 exhibit three different hydrogen binding states for a H saturated Rh(110) surface. A comparison of the exposures required to produce a certain TD state and a certain LEED pattern, respectively, revealed a close correlation between the formation of a given ordered H phase and the binding energy of the H atoms adsorbed in this phase. The first three ordered H phases, the p(1 × 3), p(1 × 2) and 1 × 3-2H give rise only to a single H binding state (denoted β) which desorbs with second-order as one would expect for a "normal" recombinative H desorption. The determined activation energy for desorption and the isosteric heat of adsorption agree to within a few kJ/mol which is evidence of a non-activated adsorption. The value of ~ 80 kJ/mol is somewhat lower than the typical adsorption energy found, e.g., with Ni(110) or Pd(110) in a similar coverage range ($E^*_{Ni(110)} \approx 84$–92 kJ/mol [3,25], $E^*_{Pd(110)} \approx 96$ kJ/mol [4,26]). It resembles more the value of the Ru(10$\bar{1}$0) surface of ~ 84 kJ/mol [24] or that of the Co(10$\bar{1}$0) surface of ~ 81 kJ/mol [27]. The absence of a multiple TD peak structure up to $\theta_H = 0.67$ indicates that, in this coverage range, the lateral interactions between the H atoms are not strong enough and predominantly attractive.

On Ni(110), Pd(110) and Co(10$\bar{1}$0) H forms low coverage phases in which the H atoms occupy threefold adsorption sites in a zig-zag configuration [4,14,27] as the centered LEED structures and recent LEED calculations [28] indicate. Interestingly, there are no such centered phases observed with Rh(110), instead the H atoms clearly form linear chains in the [1$\bar{1}$0] direction. Making use of the well-known tendency of the H atoms to occupy sites with a high coordination number [29] we propose the model illustrated in figs. 10a–10c for the first three phases. This model is a tentative one, the exact location of the adsorbed H atoms is expected from a dynamical LEED analysis which is in progress. Moreover, there are two features that deserve more attention: (i) the value of the absolute intensity of the fractional-order beams of the p(1 × 2) phase (~ 1/10 of the integer beams) and (ii) the fact that the amplitudes of the H scattered waves do not vanish at higher electron

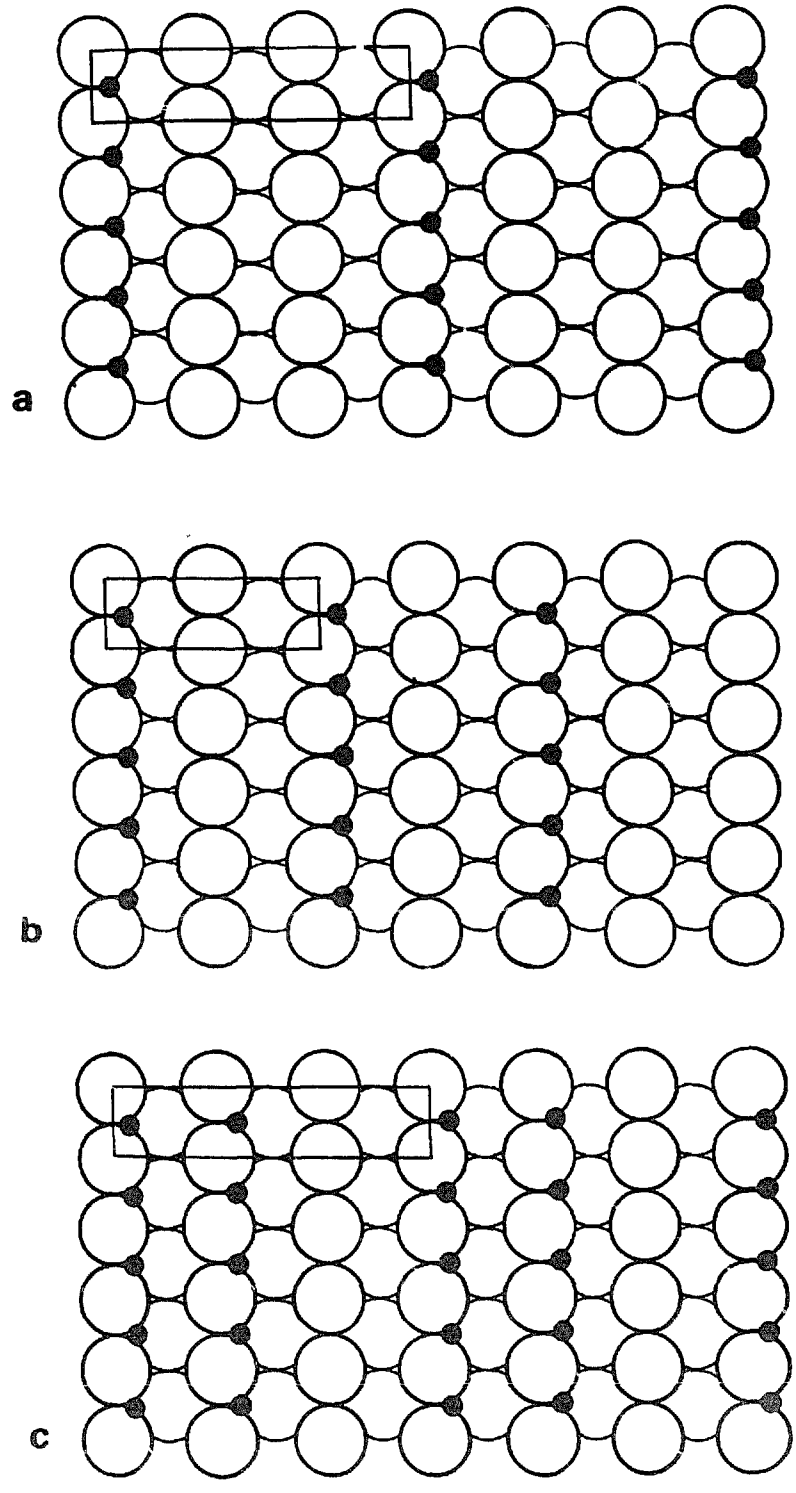


Fig. 10. Tentative structure models for the various adsorbed H (D) phases on Rh(110): (a) p(1×3) phase; (b) p(1×2) phase; (c) 1×3-2H phase.

energies (sharp fractional-order diffraction maxima are visible up to more than 300 eV beam energy). Both features suggest that there may be at least some scattering contribution also from metal atoms which have the same periodicity as the adsorbed H atoms. This apparent surface reconstruction will be discussed further below.

Similar lattice gas phases with 1×2 and 1×3 symmetry have also been reported for H adsorption on the bcc Fe(110) surface [30,31] which, of course, has a different symmetry that the fcc (110) surface, but nevertheless linear H chains are formed which tend to attract each other to form double or multiple rows in the non-primitive superstructures. An interesting point concerns the order–order transitions from a $p(1 \times 3)$ to a $p(1 \times 2)$ phase and particularly from the $p(1 \times 2)$ to the 1×3-2H phases. As the schematical drawing of fig. 11 indicates, the transition cannot simply be accomplished by a homogeneous shift of rows in [001] direction. Instead, a complicated correlated diffusion process is necessary which involves the dissolution and re-formation of entire chains. The LEED pattern in this coverage range shows clear elongations of the "extra" spots in [001] direction and provides some information about the actual phase transition mechanism: There is no complete dissolution of the lower coverage phase to a disordered structure but rather domains of higher adsorbate density are locally formed which grow at the expense of the low coverage domains. Since the streaks remain sharp in $[1\bar{1}0]$ direction there is no disorder introduced along the chains – apparently, the mobility of H in this direction is considerably lower than in the perpendicular [001] direction which points to attractive interactions in $[1\bar{1}0]$ direction. Yet it remains an interesting question how the system solves the difficult problem of the transport of matter

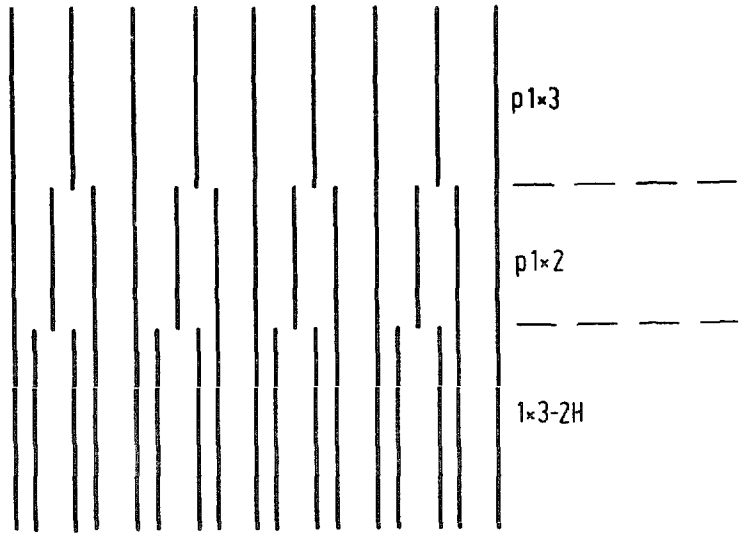


Fig. 11. Schematical drawing of the phase transitions $p(1 \times 3) \rightarrow p(1 \times 2) \rightarrow 1 \times 3$-2H. Each occupied H row (in $[1\bar{1}0]$ direction) is indicated by a full line.

when transforming from the p(1 × 2) to the 1 × 3-2H phase. The fact that at low temperatures the "extra" spots are not exactly located in the $(\frac{1}{3}, \frac{2}{3})$ positions seems to reflect these difficulties, and only temperatures well above 200 K give the H atoms enough mobility to eventually reach the energetically most stable equilibrium positions.

The structure models of fig. 10 suggest that the next phase formed after completion of the 1 × 3-2H phase should be one which all rows are filled with H atoms, i.e., a 1 × 1 phase. Quite interestingly, however, we do *not* observe such a 1 × 1 phase, instead, there is a new relatively sharp low-energy thermal desorption state which is correlated with the appearance of the second 1 × 2 LEED phase, again with sharp and comparatively intense "extra" spots (which, however, do not reach the intensity of the integer-order beams). The coverage associated with this phase is 1.5 ML. The structure model displayed in fig. 12 was developed under the assumption that no *additional* reconstruction is associated with this 1 × 2-3H phase. In order to verify the 1 × 2 periodicity one has to assume alternating single and "double" rows whereby the "double" rows consist of Rh atoms which are surrounded by two H atoms at a mutual distance of 1.7 Å. For the remaining "single" chains there are two possibilities to occupy sites, namely (i) threefold sites (this, however, implies three non-equivalent H–H distances (a, b and c)) and (ii) bridge sites whereby only two non-equivalent H–H distances are involved (c, d): Clearly, LEED structure determinations or vibrational loss measurements have to be carried

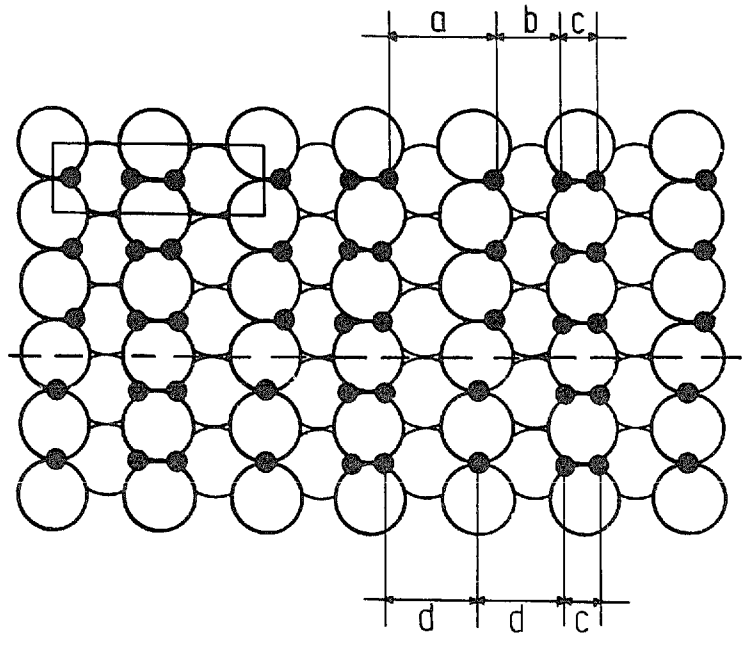


Fig. 12. Structure model of the 1 × 2-3H phase. Two possible configurations leading to different mutual H–H distances (a, b, c) are indicated in the figure (see text).

out for this phase in order to solve this question. We deduce from the α_2 TD state that the overall H binding energy in the 1×2-3H phase is reduced by -30 kJ/mol. This loss of binding energy is certainly due to repulsive interactions caused by the more closely packed H atoms on the surface.

The influence of packing density on the binding energy of the adsorbate atoms is even more pronounced for the final H structure, the 1×1-2H phase. As the structure model of fig. 13 illustrates the 1×1-2H phase now consists exclusively of double rows of H atoms with a mutual particle distance of merely 1.7 Å (distance c). All H atoms are located in equivalent threefold sites. The other (more likely) possibility is that all H atoms have to leave the "perfect" threefold sites somewhat (by ~ 0.1 Å) in [001] and [100] direction, respectively, which results in a homogeneous distribution of H atoms in [001] direction with a mutual distance of $c' \approx 1.9$ Å. The TD spectra of fig. 1 clearly reveal that a low energy α_1 state is associated with the 1×1-2H phase. Again, there is a loss in the binding energy of 20 kJ/mol with respect to the α_2 state (and 50 kJ/mol with respect to the β state). This approximate doubling of the repulsive interaction energy suggests that there is a distinct repulsive energy associated with the formation of each "close-packed" H–H pair of distance c.

A similar close-packing of H has been reported so far only for the (bcc) W(100)/H system [32]. Quite recently, we have found $\theta = 2.0$ also with the

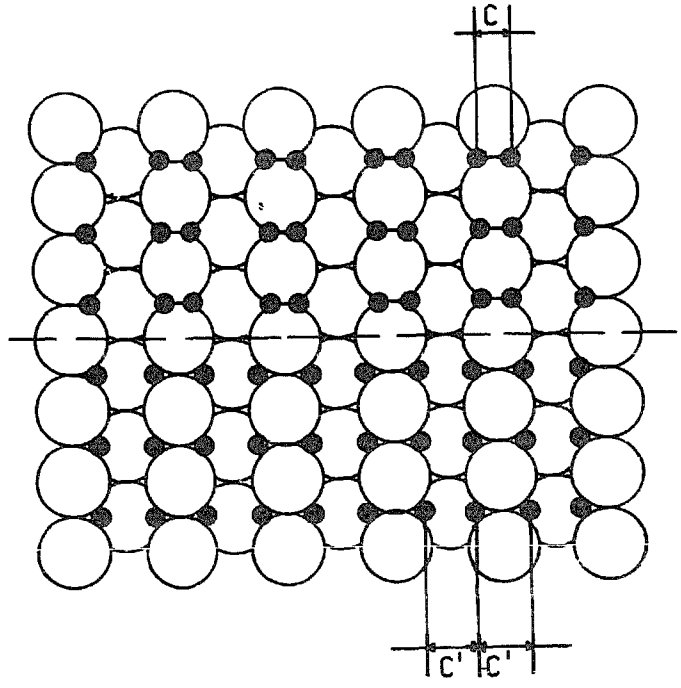


Fig. 13. Structure model of the saturation phase 1×1-2H. Assuming a uniform distribution of H atoms in [001] direction only the mutual H–H distance c' arises.

(hcp) Ru(10$\bar{1}$0)/H system. Again, the high adsorbate density caused the appearance of a low-energy binding state in this system [24].

We return to an important structural aspect, namely, the possible H-induced surface reconstruction. Numerous chemisorption studies in the past have confirmed that there is always an interplay between the interaction energies of the surface species (binding energies adsorbate–substrate atoms as well as adsorbate–adsorbate interactions) and the geometrical structure of the entire surface phase including the positions of the substrate atoms. Careful LEED structure determinations revealed that the assumption of fixed substrate atom positions is only a crude or even wrong approximation if chemically active atoms (O, N, H) adsorb on the surface [1,2,33]. Even weakly held species like hydrocarbons may eventually cause displacements of surface metal atoms [34]. The least process that can and will occur is the so-called layer relaxation whereby inherent contractions of the clean surface bond lengths are lifted in the vicinity of an adsorbed atom.

The mutual adsorbate interactions are responsible for long-range order within an adsorbate phase [35,36]. Examples for oxygen or carbon monoxide adsorption data back to the early days of LEED [37,38]. With H adsorption the situation was believed to be more complex since H is a weak electron scatterer and doubt was raised whether an ordered H adlayer could at all give rise to visible LEED "extra" spots. Therefore a 1×2 LEED superstructure observed with Ni(110)/H was attributed to a lateral displacement of Ni atoms, the so-called "pairing-row" surface reconstruction [39,40]. Subsequent thorough investigations revealed that electrons scattering off adsorbed hydrogen atoms alone could nevertheless produce LEED "extra" spots (H/Ni(111) [41] and H/Pd(111) [42] are clear examples) of, however, *weak* intensity. Since then numerous discussions were conducted whenever ordered H superstructures were observed on metal single crystal surfaces as to whether merely an ordered H adlayer or a reconstructed metal surface was responsible for the "extra" diffraction features. Two extreme cases have been distinguished. Weak "extra" intensities and unchanged LEED spectra of the integer-order beams observed with thermodynamically stable low-index planes were regarded as reflecting true H scattering whereas strong "extra" intensities along with significantly altered $I-V$ spectra seemed to indicate a H-induced surface reconstruction [43]. The H on Ni(110) system stands as a good example that this view is certainly justified. However, the present results for the Rh(110)/H system provide some evidence that the aforementioned simple view has to be modified in some detail.

Here, there are only minor alterations of the $I-V$ curves produced by adsorbed hydrogen and yet the visibility of the fractional-order beams up to high electron energies and their comparatively high intensity suggests that the substrate atoms participate in the scattering process. One may think of a slight elongation of metal atoms in z-direction, i.e., perpendicular to the surface, in

the direct vicinity of adsorbed H atoms which supports the fractional-order beam generation (up–down reconstruction). Actually, latest results from the Erlangen group reveal for the p(1 × 2) phase slight displacements of the Rh atoms (about 0.05 Å) towards the adsorbed H atoms [18]. From these current LEED calculations we expect also information about the interesting problem of whether changes in the Rh layer relaxation or up–down reconstruction are occurring step-wise in that a certain displacement is correlated with a given H adsorbate phase or if this is a more continuous process which is proportional to the overall coverage.

Another possible explanation for the changes in the surface structure of the Rh surface could be a partial solution of H in the subsurface bulk region of the Rh crystal. There are no hints whatsoever that this process occurs. Moreover, the solubility of H in Rh is endothermic and therefore very small at our adsorption temperatures [44].

4.2. Work function effects

A few remarks ought to be made about the H-induced work function change which is somewhat exceptional with the H/Rh(110) system.

With Co(10$\bar{1}$0) [27], Ni(110) [15], Ru(10$\bar{1}$0) [24] and Pd(110) [3] all an initially linear increase with H exposure or coverage is observed which persists up to $\theta_H \approx 1.0$. Only with Rh(110) a strong deviation from linearity occurs already at $\theta_H = 0.67$, i.e., when the 1 × 2-3H phase is being formed. Moreover, the maximum H-induced work function change observed with the open diagonal surfaces of the first and second row Group VIII elements is highest with Rh as the compilation in table 2 demonstrates. The exceptional high value of almost 1 eV is obviously caused by the fact that only on Rh(110) *all* H binding states exhibit a positive work function change ($\beta \approx +100$ meV, $\alpha_2 \approx +500$ meV, $\alpha_1 \approx +330$ meV). On Ru, on the other hand, only the β and α_2 state cause a positive $\Delta\varphi$ whereas α_1 and α_3 are associated with a negative work function change. Since it is very difficult to disentangle work function effects simply caused by adsorbate surface dipole moments and those originated by structural changes of the substrate (which may be more important than previously believed) it is not reasonable to argue any further about the detailed origin of the work function contributions.

Table 2

	Surface					
	Fe(211)	Co(10$\bar{1}$0)	Ni(110)	Ru(10$\bar{1}$0)	Rh(110)	Pd(110)
$\Delta\varphi$ (eV)	IP [a] [45]	+0.210	+0.510	+0.440	+0.930	+0.325

[a] IP – in preparation.

4.3. The adsorption / desorption kinetics

The structural features discussed in the foregoing section are intimately correlated with the kinetic properties of the H/Rh(110) system as will be discussed below.

In contrast to many other transition metal surfaces we find for Rh(110) the very high initial sticking probability of 0.97 ± 0.10 and a pronounced precursor state adsorption kinetics. From numerous kinetic measurements of H sticking probabilities on metal surfaces in the past one can now deduce a clear trend in that there is always a high initial sticking on atomically rough and rather a lower s_0 on atomically smooth surfaces.

Values of s_0 close to unity were reported for H on Ni(110) [15,46] Pd(110) [4], Co(10$\bar{1}$0) [27] and Ru(10$\bar{1}$0) [24], whereas on Ru(0001) [47], Pd(100) [48], Rh(100) [9] values around 0.1–0.6 and on defect-free Ni(111) [49], Ni(100) [50] or Pt(111) [51] surfaces values even less than $s_0 = 0.1$ were observed. As was shown convincingly in the group of Comsa by He beam scattering [41,52] and by Rendulic and and coworkers by means of angle-resolved thermal desorption spectroscopy [53] it is apparently the concentration of surface defects that governs the trapping and subsequent dissociation of a H_2 molecule. Whereas on ultimately defect-free Pt(111) surfaces s_0 was smaller than 10^{-3} even a slight defect concentration caused s_0 to rise by several orders of magnitude [52]. Obviously, an activation barrier exists for the trapping/dissociation process on several low-index transition metal surfaces which can only be overcome by translationally "hot" molecules. As the comparatively high values for s_0 on crystallographically more "open" surfaces demonstrate the inherent roughness of these planes provides very effective channels for trapping and subsequent dissociation. The Rh(110) is no exception here. Further information especially about the nature of the mobile precursor state can only be obtained from molecular beam experiments which could not be carried out in our apparatus. The existence of this precursor state makes it difficult to come up with conclusions about the order of the adsorption kinetics which should be 2 for purely dissociative adsorption. Similar problems arose with the system H on Ni(110) [25] and Pd(110) [4,26]. With Ni(110)/H, in particular, a H-induced surface reconstruction and a removal of a H-stabilized phase, respectively, led to curious effects in the adsorption and desorption kinetics [3]. Since reconstruction plays only a minor role with Rh(110)/H we do not observe such effects here. An interesting point, however, is the influence of the long-range order of a surface phase on the kinetics of adsorption. It is very well feasible that an impinging H_2 molecule experiences a different sticking and/or dissociation probability depending on whether it "feels" an ordered or a disordered surface layer. There is some evidence from our H_2–D_2 exchange measurements [23] that the long-range order of an adsorbate phase influences indeed the integral sticking. Apparently, it is easier for H_2 (D_2) to stick on ordered than on disordered phases.

There is one extensive previous investigation into the adsorption/desorption kinetics of H on Rh, namely the work by Yates et al. using a Rh(111) surface [9]. They found basically a $(1-\theta)^2$ proportionality of the rate of adsorption with an initial sticking probability of 0.65 ± 0.03 at 175 K; a saturation coverage of 1.58×10^{15} H atoms/cm^{-2} was obtained (H : Rh surface atom ratio of 1 : 1). A closer analysis of the kinetic data showed the existence of a mobile precursor state, similar to our case. It appears that, neglecting the higher s_0 of Rh(110)/H, the systems Rh(111)/H and Rh(110)/H exhibit some similarities in their kinetic behaviour. This holds also for the desorption process, for example, with regard to the activation energies and the order of desorption. With Rh(111), Yates et al. report on an activation energy for desorption of 78 kJ/mol for the high temperature state and on a second-order desorption kinetics. Our values are slightly higher ($E_d^* = 90$ kJ/mol), and likewise a second order is observed for the β state. The mobility of the H (D) atoms is certainly very high in the low coverage phases since these are formed instantaneously. This may change somewhat in the high coverage phases. Interestingly, the two systems Rh(111) and Rh(110) differ with regard to the influence of the precursor state on the adsorption/desorption kinetics. Rh(111) shows a stronger effect in the desorption path. The reason for this is unknown.

The kinetics of adsorption and desorption of H on Rh(110) changes entirely as one moves to the higher coverage regime: At a first glance both the α_1 and α_2 states seem to desorb via first-order kinetics. A closer inspection reveals, however, that this is only approximate since the peak shapes and widths as well as a *slight* shift of T_{max} to higher temperatures with increasing coverage (particularly for the α_1 state) indicate a more complicated thermal desorption mechanism than the simple associative desorption. The slight fractional desorption order could be explained by interaction between the H–H chains along the [1$\bar{1}$0] direction and by the fact that the desorption from the "ends" of the chains requires somewhat less activation energy than desorption out of the middle of a chain. Here again we have an interesting influence of the adsorbate structure on the kinetics of desorption, whereby it is established from our LEED measurements that the desorption giving rise to the α_1 and α_2 states occurs from the *ordered* 1×1-2H and 1×2-3H phases, respectively (their order–disorder transition temperatures are higher than their desorption temperatures).

Another interesting observation concerns the kinetics of the formation of the final 1×1-2H phase. It is possible by a careful partial heating to desorb only the population of the α_1 state. Postadsorption of hydrogen at the typical adsorption temperature of 85 K (to which the surface was cooled down again after the partial desorption) led to the formation of the 1×1-2H phase only after a much longer period of time than was required in the initial experiment. This is illustrated in fig. 5. Apparently even the slight heating to only 160 K

introduces enough disorder in the 1×2-3H phase that the subsequent formation of the 1×1-2H phase is perturbed. Also interesting is that this effect is less pronounced (although still significant) if one uses D_2 instead of H_2.

A final remark ought to be made about the interesting high saturation coverage found with Rh(110)/H (which could not be reached with Rh(111) between 100 and 500 K [9]). Taking the closest Rh–Rh distances of the Rh(111) surface, i.e., the distance of two adjacent threefold coordinated sites, which is 1.54 Å, it is clear that this is too close for all these sites to become occupied by H atoms. Therefore no 1×1 phase can be formed on Rh(111). This is somewhat equivalent to the situation with Ru(0001) and Ru(10$\bar{1}$0)/H. For the smooth (0001) plane Lindroos et al. [55] report on a saturation coverage of 1.6×10^{15} H atoms/cm^2 ($\theta_H = 1.0$), whereas we find on Ru(10$\bar{1}$0) a $\theta_{H,max}$ of 2.0 [24] (whereby, interestingly, the H atoms very likely are also arranged in linear chains). On Pd(110), the other similar surface, it could be that $\theta_H \neq 2$ could be formed if there were not a progressive surface reconstruction occurring at $\theta_H = 1.5$ which leads to row-pairing [56] and a dissolution of H in the bulk [57].

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